

Creator Education Platform

Full Backend Architecture Blueprint using Django REST Framework

This document defines the backend architecture for a Creator Education Platform. Creators can publish courses, offer subscriptions, and interact with followers while users can enroll in courses and subscribe to creators. The project is designed to practice advanced backend engineering concepts including complex database modeling, subscription billing, event driven systems, analytics, background processing with Celery, and caching using Redis.

Technology Stack

- Django
- Django REST Framework
- PostgreSQL
- Redis (Caching + Celery broker)
- Celery (Background jobs)
- JWT Authentication
- Django Filters
- Pytest or DRF testing tools

Core System Modules

- Authentication System
- Creator Profile System
- Course Management System
- Lesson Content System
- Subscription Billing System
- Payment Processing System
- Follower and Engagement System
- Notification System
- Analytics and Reporting System

Core Database Models

- User
- CreatorProfile
- Follower
- Course
- Module
- Lesson
- Enrollment
- LessonProgress
- Certificate
- Product
- Plan
- Subscription
- Invoice
- Payment
- Coupon
- Comment
- Like
- Notification
- ActivityLog
- CreatorAnalytics
- RevenueReport

Background Tasks (Celery)

- Generate certificates after course completion
- Send subscription renewal reminders
- Generate monthly creator revenue reports
- Retry failed payments
- Send notification emails
- Clean expired subscriptions

Testing Strategy

- Authentication tests
- Permission tests
- Course creation tests
- Enrollment workflow tests
- Subscription billing tests
- Payment processing tests
- Notification trigger tests